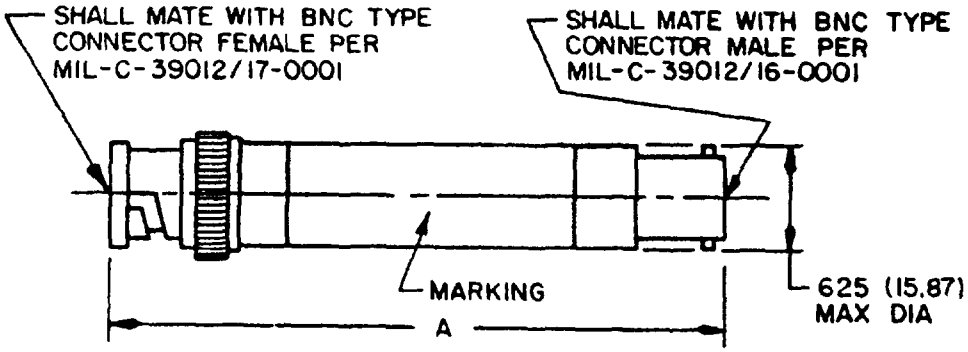
MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (BNC CONNECTORS),

FREQUENCY RANGE: DC TO 4.5 GHz, CLASS III, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.



NOTES
1 Dimensions are in inches.
2 Metric equivalents are given for general information only.
3 Metric equivalents are in parentheses

FIGURE 1 Dimensions and configuration

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ENGINEERING DATA:

Class: III.

Operating frequency range: DC to 4.5 GHz.

Test frequencies for VSWR and attenuation measurements: dc, 1, 2, 3, and 4.5 GHz.

Basic test frequency: 2 GHz.

Operating temperature range: -55°C to 125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power rating: See table I.

VSWR: See table I.

Temperature coefficient: .0001dB/dB/°C.

Part number: M3933/19- (and dash number from table I).

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB			Maximum power input @ 25°C		VSWR maximum 1/				Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation		AV (W) 2/ (continuously)	Peak (kW)	dc to 1.0 GHz	1.0 to 2.0 GHz	2.0 to 3.0 GHz	3.0 to 4.5 GHz		
		DC	DC to 4.5 GHz								
01 N S	1	±0.1	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
02 N S	2	±0.1	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
03 N S	3	±0.1	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
04 N S	4	±0.1	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
05 N S	5	±0.1	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
06 N S	6	±0.1	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
07 N S	8	±0.2	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
08 N S	10	±0.2	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
09 N S	11	±0.2	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
10 N S	12	±0.2	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
11 N S	15	±0.2	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
12 N S	20	±0.2	±0.5	2	0.5	1.10 1	1.15·1	1.20:1	1.25:1	4	2.7 (68.6)

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB			Maximum power input @ 25°C		VSWR maximum 1/				Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation				dc to 1.0 GHz	1.0 to 2.0 GHz	2.0 to 3.0 GHz	3.0 to 4.5 GHz		
		DC	DC to 4.5 GHz	AV (W) 2/ (continuously)	Peak (kW)						
13 N S	25	±0.4	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
14 N S	30	±0.4	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
15 N S	40	±0.4	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
16 N S	50	±0.5	±1.2	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
17 N S	60	±0.5	±1.4	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
18 N S	13	±0.2	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)

1/ VSWR value is for both end.

2/ Power input is derated linearly to 0.5 watts at +125°C.

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Note: Attenuators covered by this specification sheet are substitutable for the manufacturer's or user's part numbers specified in table II. This information in no way implies that manufacturer's part numbers is suitable as a substitute for the military part numbers.

TABLE II. Substitution information.

Military part number M3933/19-	Manufacturer's code	Manufacturer's part number	NSN
01	23042	FP50-10BBNC	5985-01-141-4322
01	94117	228044P1	---
01	00929	AA-D98	---
01	24930	G300009S2	---
01	99899	755-1	5985-01-028-0357
01	18203	A401B	5985-01-028-0357
01	00929	AA-01B	---
02	94117	228044P2	---
02	00929	AA-089	---
02	23042	FP-50-2BNC	5985-01-136-6947
02	20418	614175-1	5985-01-136-6947
02	00929	AA-02B	5985-00-816-9360
03	24930	G300009S3	5985-00-030-8920
03	99899	755-3	5985-00-030-8920
03	18203	A403B	---
03	00929	AA-03B	5985-00-747-9667
03	00929	AB-03B	5985-00-899-2153
03	23042	FP50-3BNC	---
03	94117	228044P3	---
03	00929	AA-D90	---
04	23042	FP50-4BNC	5985-00-874-0822
04	94117	228044P4	---
04	00929	AA-D91	---
04	00929	AB-04B	5985-00-021-6449
05	23042	FP50-5BNC	5985-00-874-0826
05	94117	228044P5	---
05	00929	AA-D92	---
06	91417	117566-001	5985-00-413-1251
06	23042	FP50-6BNC	5985-00-413-1251
06	94117	228044P6	---
06	00929	AA-D93	---
06	00929	AA-06B	5985-00-882-8440
06	00929	AB-06B	5985-00-945-9991
06	23042	FP50-6BNC	5985-00-022-3506
06	06845	2073453-704	5985-00-022-3506
06	24930	G300009S4	---
06	06074	D1980-6	5985-00-022-2892
06	19156	B001241-2	5985-00-022-2892
06	99899	755-6	5985-00-022-2892
06	18203	A406B	---

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Military part number M3933/19-	Manufacturer's code	Manufacturer's part number	NSN
07	94117	228044P7	---
07	00929	AA-D94	---
07	34078	313-8DB	5985-01-105-6184
07	23042	FP50-8BNC	5985-00-576-9553
07	00929	AB-08B	5985-00-187-7815
07	00929	AA-089	5985-00-592-0372
08	94117	228044P8	---
08	00929	AA-095	---
08	03538	7844413	---
08	23042	FP50-10BNC	5985-00-689-1923
08	24930	G300009S1	---
08	18203	A410B	5985-00-784-9276
08	09017	704708-10	5985-00-784-9476
08	99899	755B-10	5985-00-686-8315
08	18876	9144037	5985-00-686-8315
08	00929	AA-10B	5985-00-802-7359
08	80249	PP2113	5985-00-802-7359
08	22424	600-10-2	5985-00-802-7359
08	28527	2210962	5985-00-802-7359
08	87557	9217C0009-6	5985-00-802-7359
08	00929	AA-10B	5985-01-089-3229
08	23042	FP50-10	5985-01-089-3229
08	23042	FP50-10B BNC	5985-01-141-4322
09	23042	FP50-11BNC	5985-01-256-5071
10	23042	FP50-12BNC	5985-01-084-5729
10	06845	2073453-0707	5985-01-084-5729
10	00929	AA-12B	5985-00-954-5469
10	79500	342C842H05	5985-00-954-5469
10	94117	228044P9	---
10	00929	AA-D96	---
10	00929	A9-12B	5985-00-102-8129
10	23042	FP50-12	5985-01-099-3228
11	94117	228044P10	---
11	00929	AA-D97	---
11	00929	AA-15B	5985-00-755-2513
11	99899	755-15	5985-01-009-5476
11	23042	FP50-15BNC	5985-00-477-8163
11	23042	FP50-15BNC	5985-01-090-7339
12	00929	AA-20B	5985-00-845-4395
12	18203	A4203	5985-00-845-4395
12	00929	AA-20B	5985-01-089-6160
12	12436	666193-655	5985-00-845-4395
12	99899	755-20	5985-00-845-4395
12	99899	755-20	5985-01-009-5477
12	23042	FP50-20BNC	5985-01-137-4237
12	23042	FP50-20	5985-01-089-6160

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TABLE II. Substitution information - Continued.

Military part number M3933/19-	Manufacturer's code	Manufacturer's part number	NSN
12	24930	G300009S5	---
12	94117	228044P11	---
12	00929	AA-D98	---
13	12436	810000-164	5985-00-959-8316
13	00929	AA-25B	5985-00-959-8316
13	94117	228044P12	---
13	00929	AA-D99	---
14	00929	AA-30B	5985-00-911-0545
14	94117	228044P13	---
14	00929	AA-E01	---
14	18203	A430B	5985-01-102-8569
14	23042	FP50-30	5985-01-089-3230
14	23042	FP50-30BNC	5985-01-079-9120
14	00929	AA-30B	5985-01-089-3230
15	18203	A440B	5985-01-491-6503
15	94117	228044P14	---
15	00929	AA-E02	---
15	00929	AB-40B	5985-00-896-9970
15	00929	AA-40	5985-01-093-0259
15	23042	FP50-40	5985-01-093-0259
16	94117	228044P15	---
16	00929	AA-E03	---
17	94117	228044P16	---
17	00929	AA-E04	---
18	00929	AA-13B	5985-01-135-2076

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Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC, SH, AS

Agent:
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Preparing activity:
Navy - EC

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